

Dear HP Enterprise Hiring Team,

I am excited to apply for the FE Simulation Engineer position at Hewlett Packard Enterprise. With a Ph.D. in Computational Mechanics and extensive experience developing finite element models and simulation frameworks for reliability analysis in microelectronics and advanced materials systems, I am eager to contribute to HPE's simulation-driven design and quality initiatives.

During my doctoral research on adhesion in microelectronics, I designed and implemented an integrated experimental-numerical framework combining finite element analysis (MSC Marc/Mentat), custom instrumentation, and full-field Digital Image Correlation (IDIC) to characterize mechanical behavior, interfacial adhesion, and failure modes in multi-layer microelectronic systems. This work involved material modeling, constitutive parameter identification, and providing engineering insights into design variations and reliability risks — skills directly applicable to HPE's needs in component and system-level reliability evaluation.

I am particularly drawn to HPE's focus on using simulation early in the design cycle to improve product quality and accelerate innovation. I am eager to collaborate with cross-functional teams to identify risks, optimize designs, and support the development of reliable, high-performance computing hardware.

I would welcome the opportunity to discuss how my background in finite element simulation, microelectronics mechanics, and reliability analysis can add value to HPE.

Sincerely,

Andre Ruybalid

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